

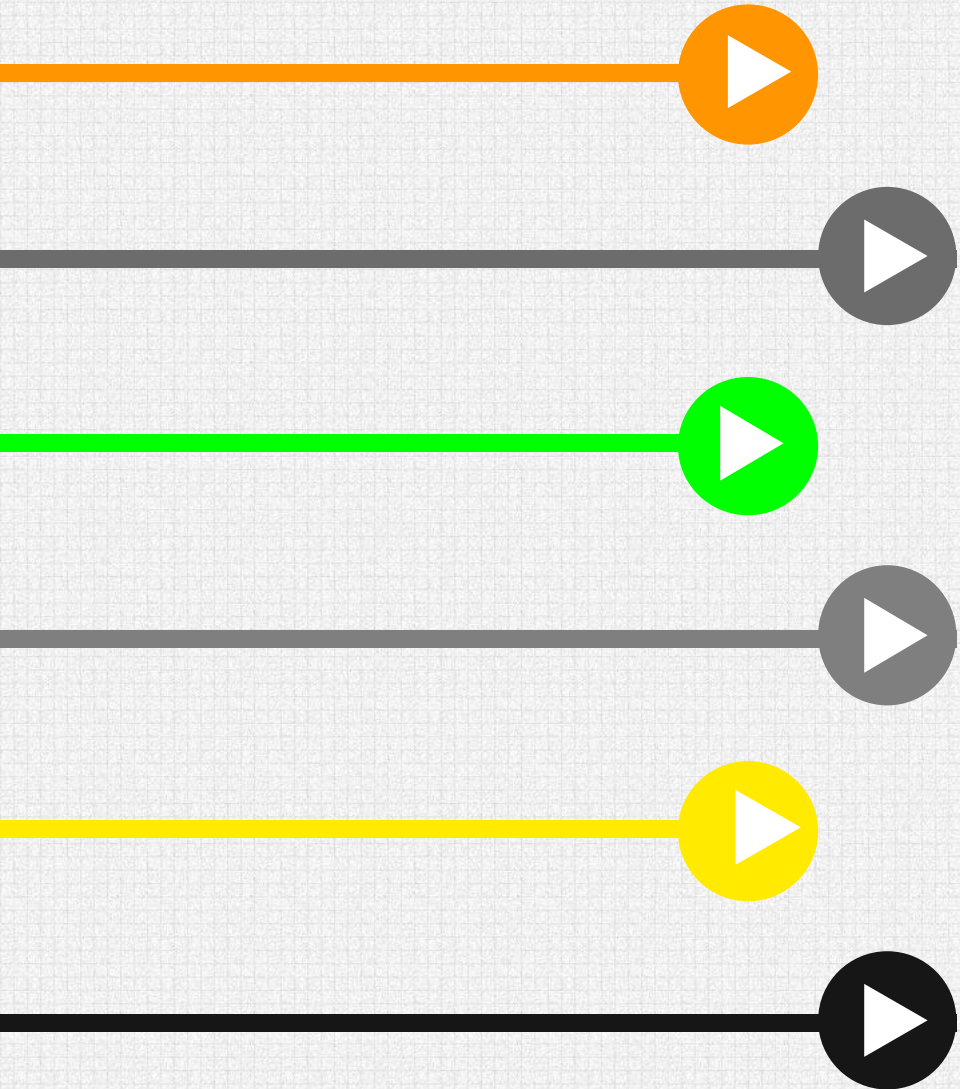


## DIGITAL IDENTITY & ACCESS MANAGEMENT

13 March 2017

**DIGITAL AMBASSADOR NETWORK**  
EALA Digital Security & Privacy

# AGENDA



Introduction

Identity and Access Management Components  
and services

Role of IAM in the digital age

Operational and Business Benefits

Digital Identity and access Management  
Technologies

Appendix



# Introduction



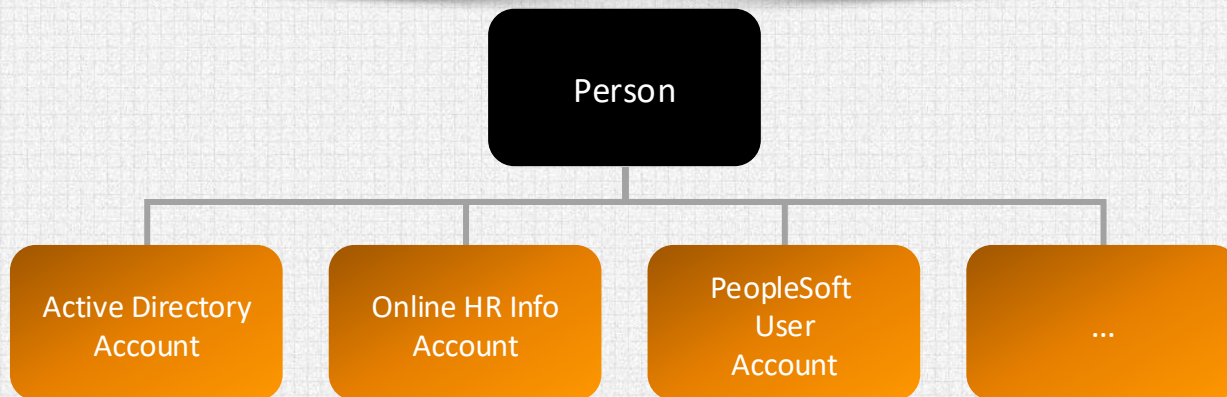
# INTRODUCTION – DIGITAL IDENTITY AND ACCESS MANAGEMENT

## Objectives

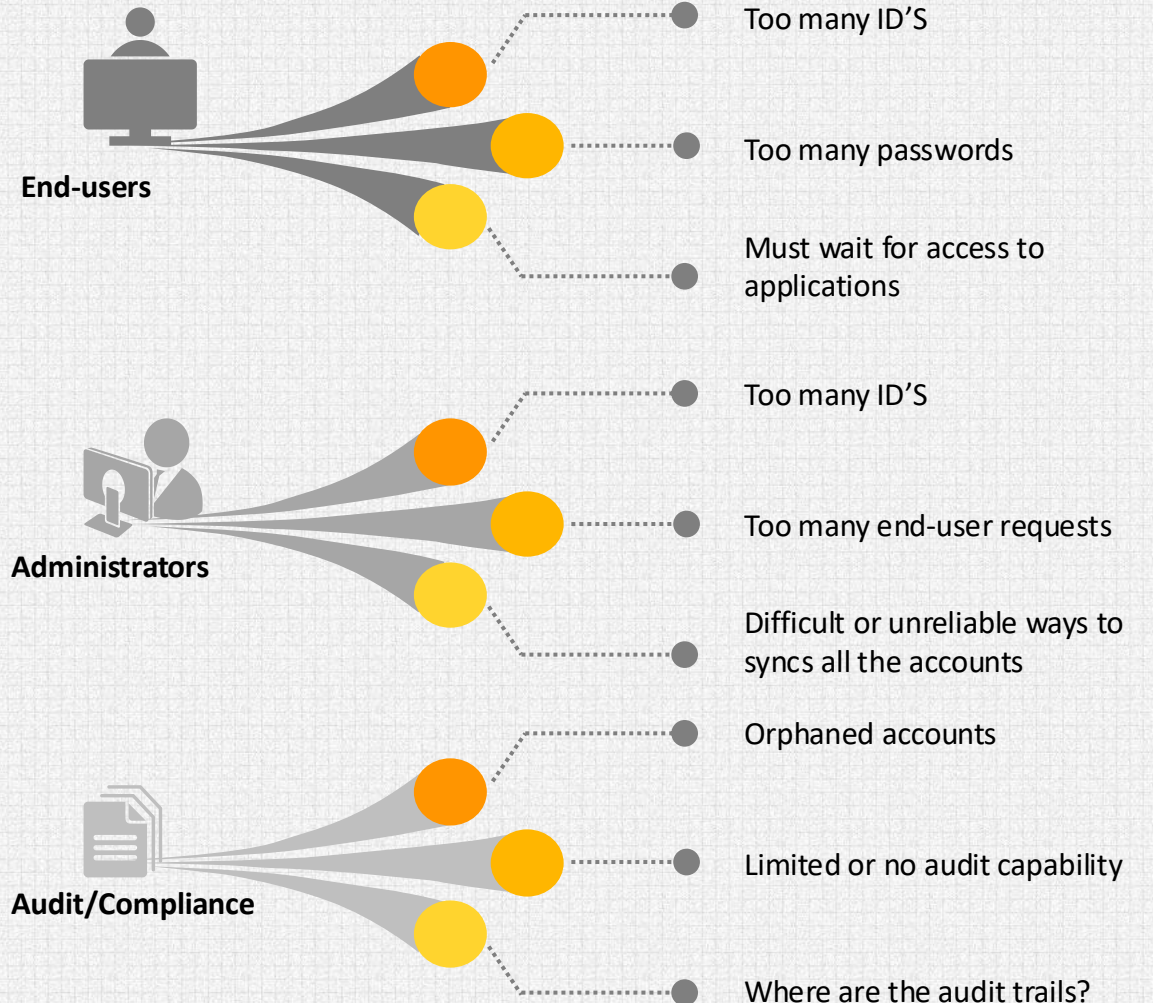
- Find a common background for discussing IAM
- Discuss problems and opportunities in the field
- Introduce terminology
- Highlight a possible future direction

### The question is about Identity. Who am I and Who are you?

- Networks use multiple identity systems
- The Internet is no better
- Users get confused with all of these IDs
- Management and audit has difficulty keeping track of all these IDs



## The real Impact





# MULTIPLE CONTEXTS AND NEED FOR HAVING AN IAM SOLUTION IN THE DIGITAL ENVIRONMENT



## Need for having an IAM system

### Regulation and Compliance

- SOX, HIPAA, GLB, GDPR

### Increasing Threats

- Identity theft
- Exposure of confidential info

### Maintenance Costs

- The average employee needs access to 16 applications
- Companies spend an estimated \$20-30 user/year for password resets

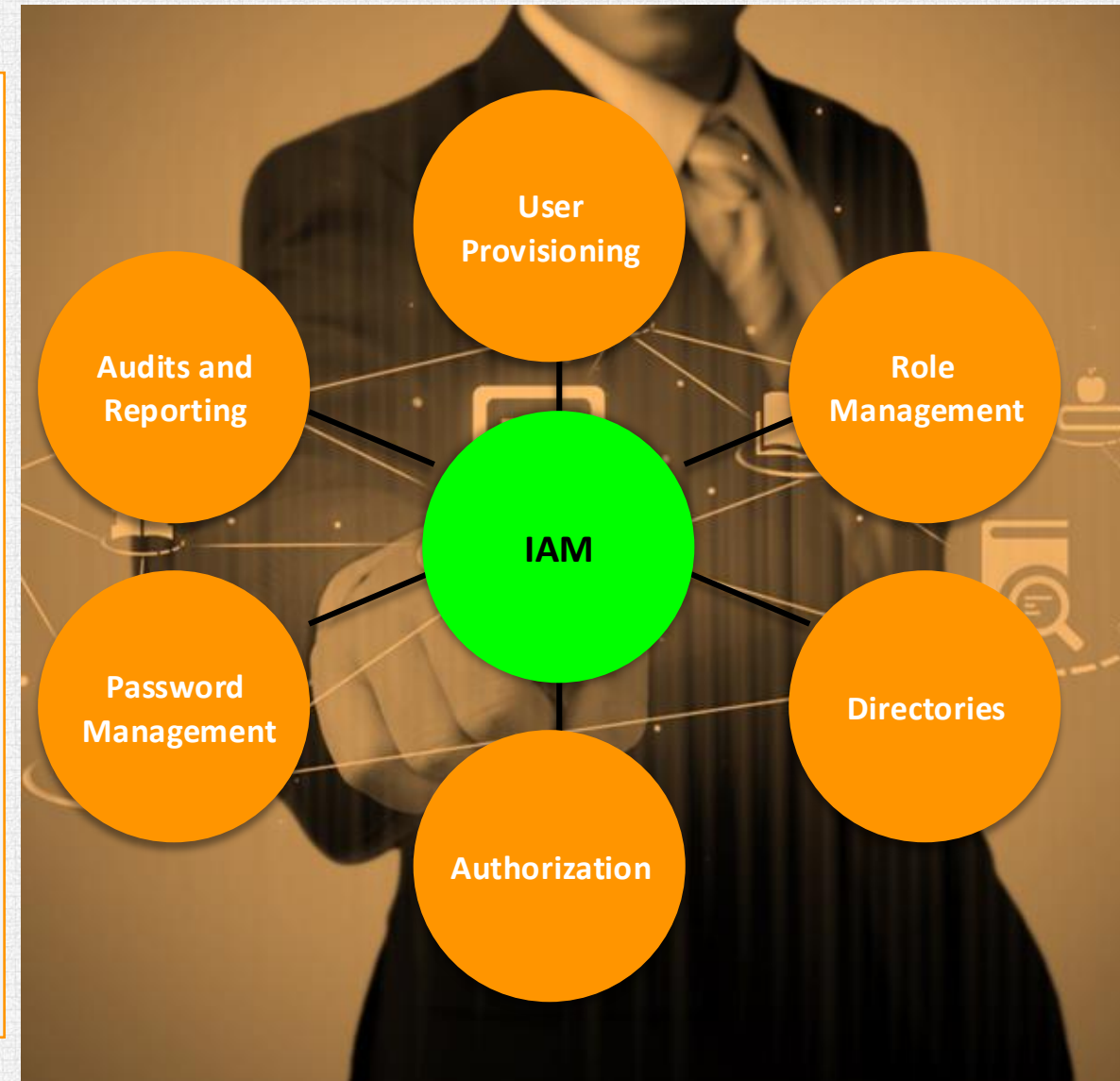


# **Identity and Access Management Components and services**



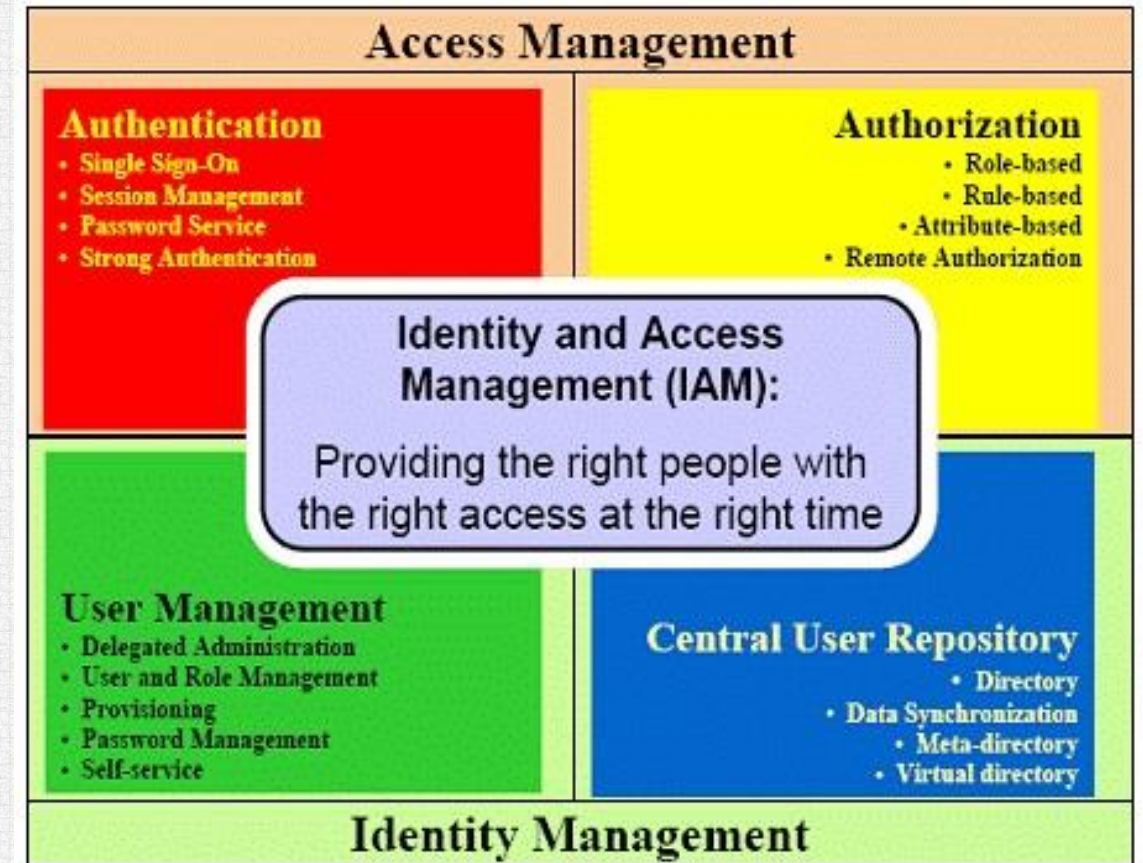
# IDENTITY AND ACCESS MANAGEMENT (IAM) – MULTI FUNCTIONAL ASPECTS

- **User Provisioning** - Manages (Create/Modify/Terminate) the identities like person, device, application etc. which interact with an organization.
- **Role Management** - Maintain right information for each identity based on trusted source.
- **Directories** - Provides appropriate level of identity trust between the identities.
- **Authorization** - Ensures that the right people gain access to the right materials and records at the right time as well as making it safe, secure, and simple.
- **Password Management** – Provides the option to manage and enforce password rules. Also functions as a Authentication mechanism
- **Audits and reporting** – Provides a one-stop shop by generating logs which could be used for the purpose of audits and assessments



# IDENTITY AND ACCESS MANAGEMENT COMPONENTS

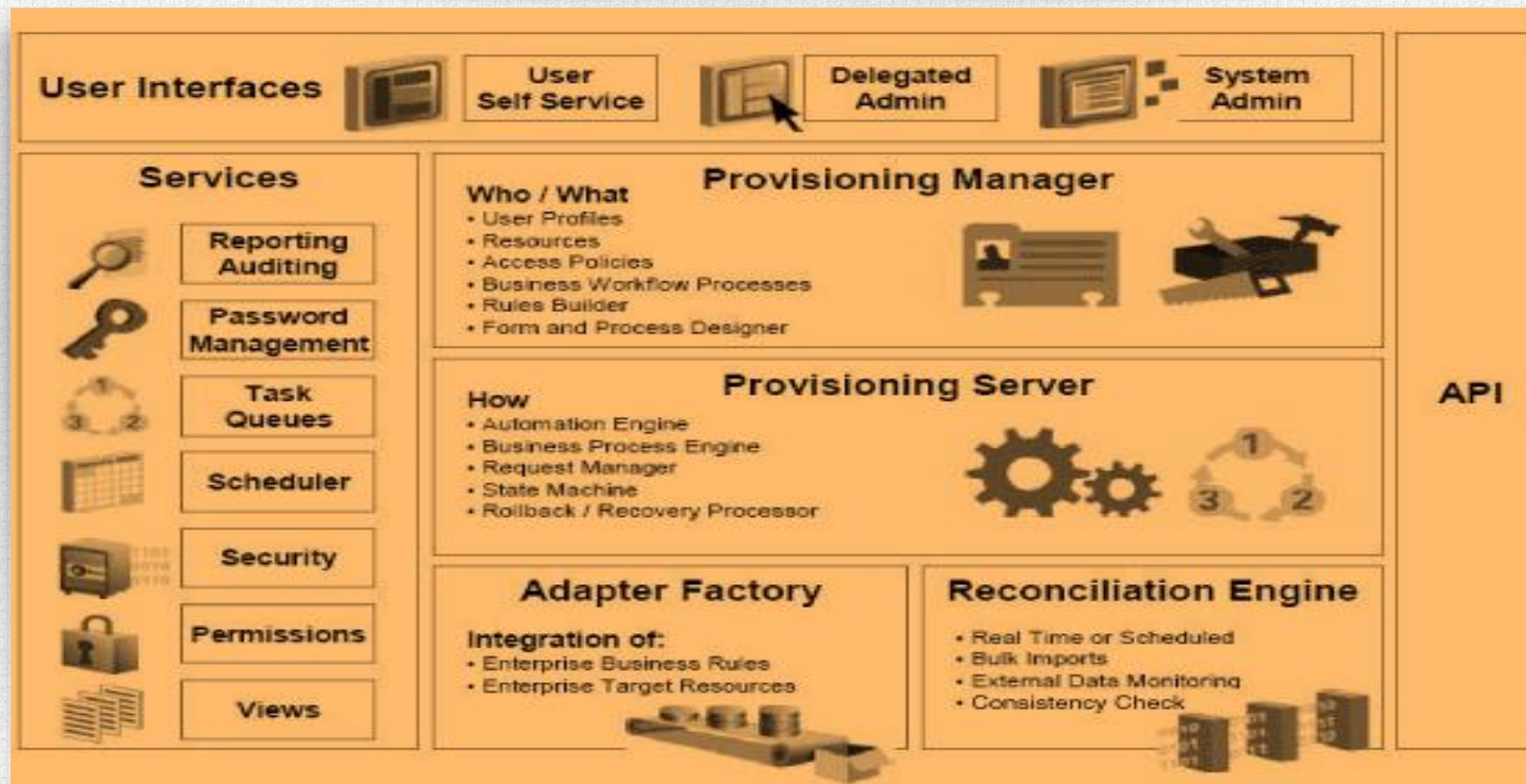
- **Identity management (IDM)** - describes the management of individual identities, their authentication, authorization, roles and privileges/permissions within or across system and enterprise boundaries with the goal of increasing security and productivity while decreasing cost, downtime and repetitive tasks.
- **Access management** is the process of granting authorised users the right to use a service, while preventing access to non-authorised users. Access management provides the right for users to be able to use a service or group of services.





# WHAT IS IDENTITY MANAGEMENT?

**Identity management (IDM)** describes the management of individual identities, their authentication, authorization, roles and privileges/permissions within or across system and enterprise boundaries with the goal of increasing security and productivity while decreasing cost, downtime and repetitive tasks.





# IDENTITY MANAGEMENT - SERVICES

## User provisioning

- User new Hire, rehire, termination, suspension functionality with approval capability.
- Applications includes Active Directory, Exchange, Siebel, Remedy, ERP, GRC and OBIEE/BI
- Auto provisioning, Manual Provisioning, Auto Access grant, Additional Access Request (Approval capability) etc.
- Retain existing accounts and link with single accounts using common identifier.

## Password Management

- Change Password, Reset Password and Password recovery
- Password Policy
- Centralization of user credentials

## Reporting Auditing

- Select and view reports from a predefined list available by-default from product
- Filter report information.
- View reports on-screen in the desired format.
- Provide interactive reports.





# IDENTITY MANAGEMENT – SERVICES CONT..

## Schedulers

- Enables you to schedule jobs that automatically run predefined scheduled tasks at the specified time.
- Creating custom scheduled tasks according to your requirements if none of the predefined scheduled tasks fit.

## Reconciliation

- Reconciliation is the process of synchronizing account between the managed resource and the Identity Manager Server.
- Reconciliation engine takes care of finding the matching reconciliation rule against the resource and applies the action defined like link account or update account.

## Adapters

- Adapters are used as a connection medium between the identity manager server and the endpoint.
- Adapters extends the internal logic and functionality of Identity Manager.

## API

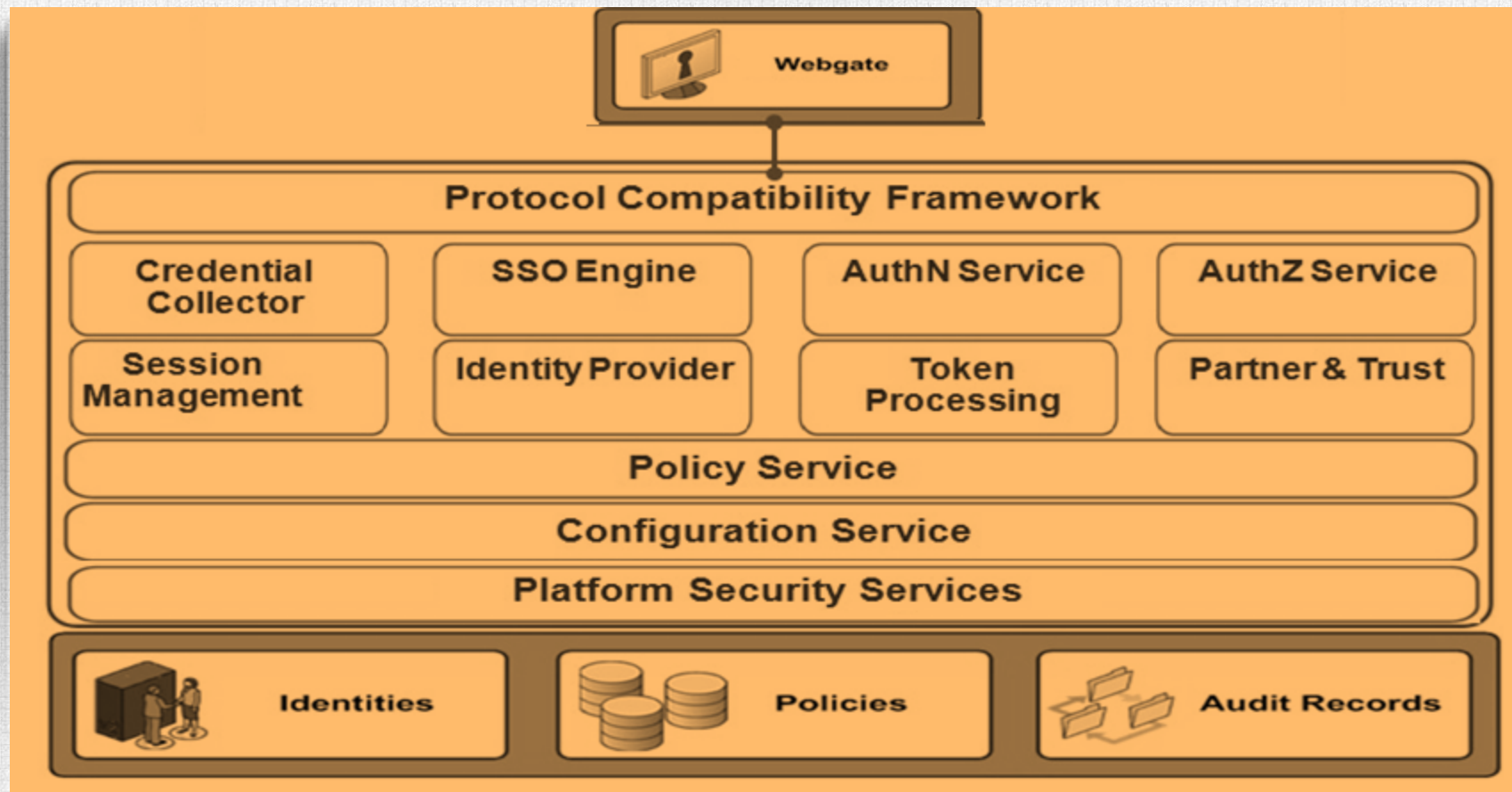
- Application Programming Interface (API) exposes services based on Plain Old Java Objects (POJO) to interact with Identity Manager. API can be used for building clients and integrating third-party products with the Identity Manager platform.





# WHAT IS ACCESS MANAGEMENT?

**Access management** is the process of granting authorised users the right to use a service, while preventing access to non-authorised users. Access management provides the right for users to be able to use a service or group of services.





# ACCESS MANAGEMENT - SERVICES

## Webgates

- Webgate is typically deployed on web server to protect the applications being served by the particular web server.
- WebGate is a web-server plug-in for Access Manager that intercepts HTTP(S) requests and forwards them to the Access Server for authentication and authorization.

## Single Sign-On

- Eliminate the need for additional or different logins to access other applications during the same session.
- Access Manager provides single sign-on (SSO) through a common SSO Engine that provides consistent service across multiple domains.
- Authentication supports key modules like LDAP, Kerberos, Certificate (X.509)

## Session Management

- Tracks active user sessions using a high performance distributed cache
- Admin can specify Session Lifetime & Idle Timeout globally
- Admin can limit the number of concurrent sessions a user can have at one time.





# ACCESS MANAGEMENT – SERVICES CONT..

## Policy Service

- Policy store is a data repository which stores the policy data, including authentication modules and schemes, Application Domains, and policies.
- Password Management data includes password policy type for each configured User Identity store as well as the policy that governs password requirements, expiry, notification, account lockout.

## Identity Store

- System Identity Store is primarily used to identify Administrators who can login and perform operations and associated information like group membership.
- User Identity Store is designated is primarily configured with different directory services to validate the user access during authentication and authorization. These identity store are typically configured with LDAP or database repositories.

## Fraud Detection

- Provides a unique combination of knowledge-based authentication with registration, answer, and fuzzy logic for the Oracle Identity and Access Management Suite
- Delivers real-time fraud detection and proactively prevents fraud.





# **Role of IAM in the digital age**



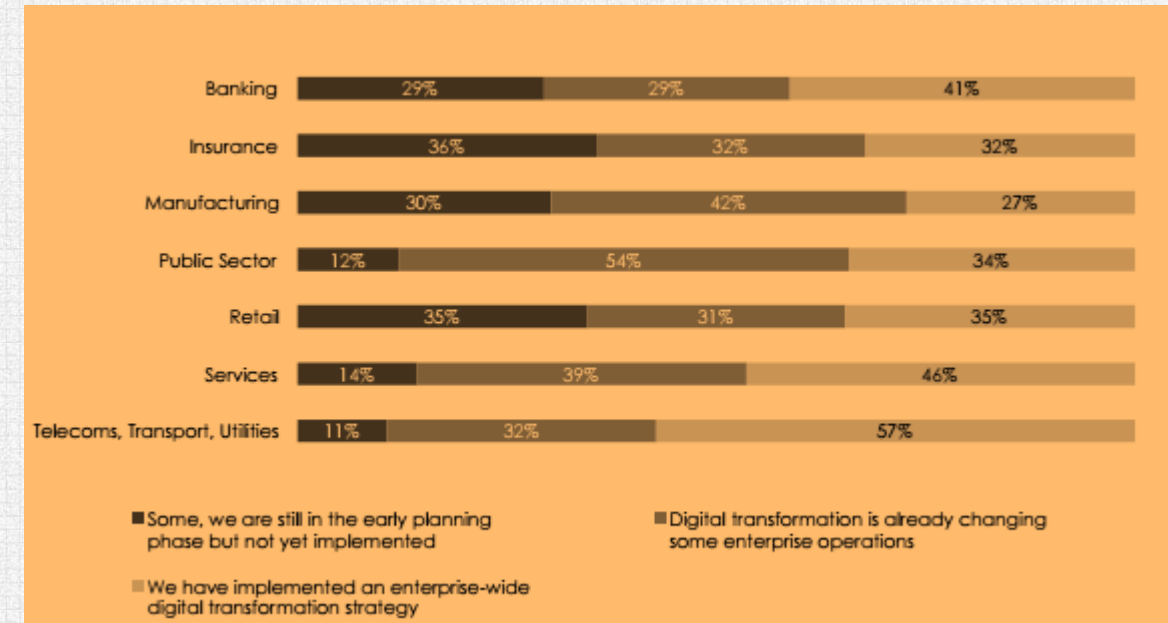
# THE ROLE OF IAM IN THE DIGITAL AGE

Embracing digital transformation across businesses to improve competitiveness, gain efficiencies or get closer to customers is becoming a challenge. More than Buzzword digital transformation is happening across the world

The heart of securing the identities is at the heart of keeping the organizations secure

## Digital Identity and Access Management Systems

- Manages the services from cloud services, mobile, Certificates and IOT services.
- Designed across multiple industries like Healthcare, Finance, Retail, Telecom, Government etc.
- Implements new security mechanisms to enable high standard of authentication and authorization.



**According to KPMG survey in 2016**



# **Operational and Business Benefits**



# OPERATIONAL BENEFITS

## User Provisioning

- Reduce account management tasks for application owners and custodians
- Improve security and provide liability to the customers.

## Password recovery

- Self-service utility to reset password by the customers using OTP, Secret Q&A.
- Reduce Help desk calls for password management

## Entitlement

- Automatic grant access based on user position
- Restrict unauthorized access to application

## Approvals

- Grant access through approval process

## Single Sign-On

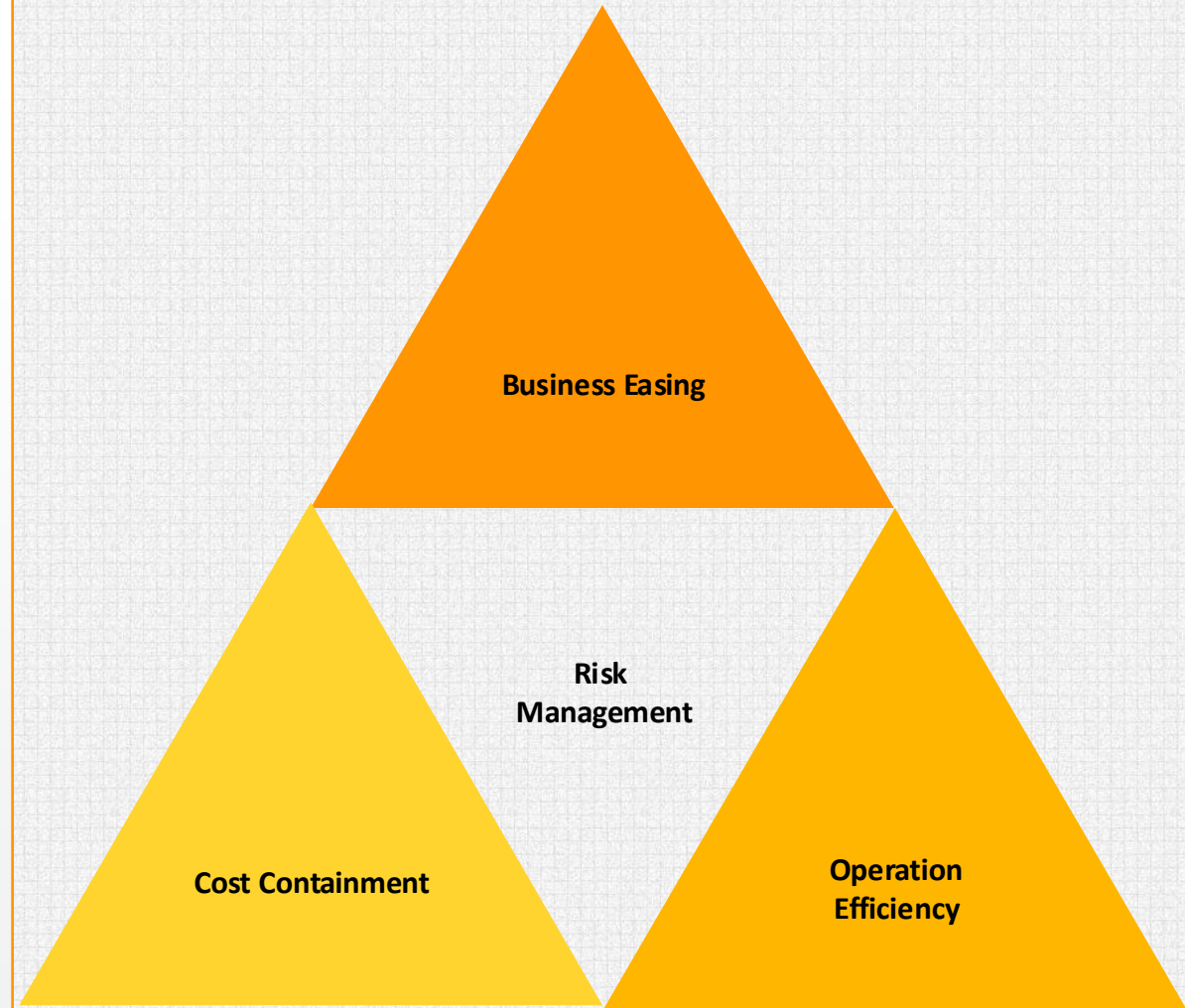
- Secure way of accessing applications (SSO)





# BUSINESS BENEFITS

- **Cost Reduction** – Self-service, automation, and improved resource allocation visibility will help reduce a company's cost inside and outside of IT.
- **Enhanced Productivity** – Automation, workflow, and self-service will result in more efficient processes, delivering quicker response times and allowing staff to focus on higher-leverage tasks.
- **Improved User Experience** – Single sign-on and federated identity will reduce user frustration and improve their usage of key applications.
- **Risk Reduction and Security Improvement** – Automation and centralized administration will allow a company to align access with job functions, ensure security policies are consistently applied, and enhance the reliability of security processes.
- **Regulatory Compliance Assurance** – Centralization and automation will allow for automated audit reviews, enhanced compliance-related tracking of user activity, expedient access certification, and improved confidence in IAM processes.





# BUSINESS BENEFITS – USE CASE

| Process                                                                                       | Manual Process         | IAM Process            | Benefit         |
|-----------------------------------------------------------------------------------------------|------------------------|------------------------|-----------------|
| Number of New hire/Rehire tickets – Application owners (Siebel, Remedy, EBS, AD/Exchange, BI) | 100X6=600 per month    | 10 per month (approx.) | - 590 per/month |
| Number of New hire/Rehire tickets – Service desk                                              | 100 per month          | 5 per month (approx.)  | - 95 per/month  |
| Employee On boarding                                                                          | 16 hours               | 15 minutes             | 15h 45m         |
| Employee Exit Process                                                                         | 8 hours                | 15 minutes             | 7h 45m          |
| Employee Transfer                                                                             | 54 hours               | 15 minutes             | 53h 45m         |
| Employee Rehire process                                                                       | 48 Hours               | 15 minutes             | 47h 45m         |
| Employee suspension                                                                           | 8 hours                | 15 minutes             | 7h 45m          |
| <b>Other benefits</b>                                                                         |                        |                        |                 |
| Audit compliance                                                                              | Manual                 | Automated              | Yes             |
| Enterprise Level Application Authorization Reports                                            | Effort required        | Automated              | Yes             |
| Login/Logout                                                                                  | Multiple Logins/Logout | Single Login/Logout    | Yes             |



# **Digital Identity and access Management Technologies**



# DIGITAL IDENTITY AND ACCESS MANAGEMENT TECHNOLOGIES

Digital identity and access management (DIAM) systems are essential to security frameworks for their ability to rapidly and consistently confirm identities and to control individuals access to resources and services.

- **Connectors**
- **Federation**
- **Web-Services**

**Identity Manager connectors** are used to integrate with external, identity-aware applications. The Identity Connectors Framework and Toolkit is built to develop of Connectors. Connectors provide a consistent generic layer between applications and target resources. Following are the features of connectors:

- **Provisioning**
- **Reconciliation**
- **Password Synchronization**
- **Data Validation**

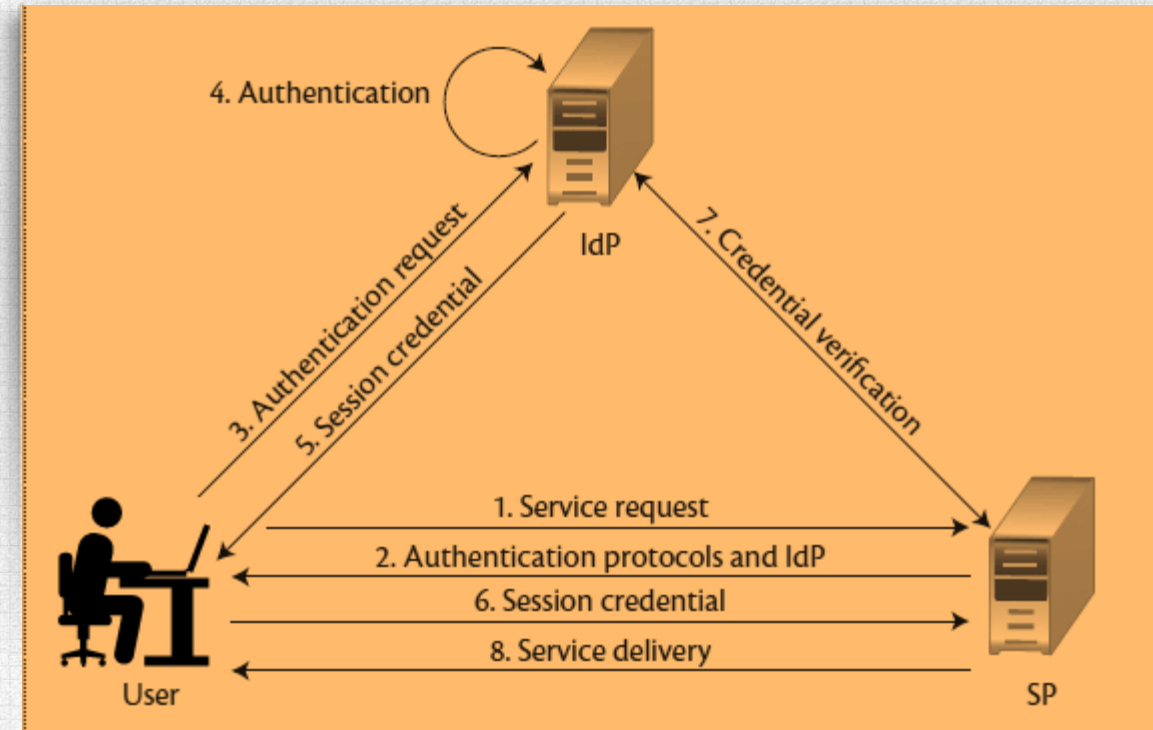




# DIGITAL IDENTITY AND ACCESS MANAGEMENT TECHNOLOGIES CONT..

## Federated Identity Management

- Enables identity information to be developed and shared among several entities and across trust domains.
- Permit identity attributes to be flow from one trusted identifying and authenticating entity to another for authentication, authorization
- Provides “single sign-on” convenience and efficiencies to identified individuals, identity providers and relying parties.





# DIGITAL IDENTITY AND ACCESS MANAGEMENT TECHNOLOGIES CONT..

**Web services** are designed to define and implement in heterogeneous environments including authentication, authorization, message encryption and decryption, signature generation and validation.

Web services in Identity and Access Management are mainly used in cloud services, federation and fusion middleware (SOA).

Identity Manager Web services are accessed using SOAP messages over an HTTP connection.

Web services in federation provides trust relationships between identity providers and service providers.

Authentication and attribute based access controls provides improved security to cloud web services.





# DIGITAL IDENTITY AND ACCESS MANAGEMENT SERVICES

**Digital identity and access management (DIAM)** systems are essential to security frameworks for their ability to rapidly and consistently confirm identities and to control individuals access to resources and services.

- Identity Governance
- Consumer Identity
- Future services
  - Mobile IDM
  - Cloud IDM
  - Biometrics





# DIGITAL IDENTITY AND ACCESS MANAGEMENT SERVICES CONT..

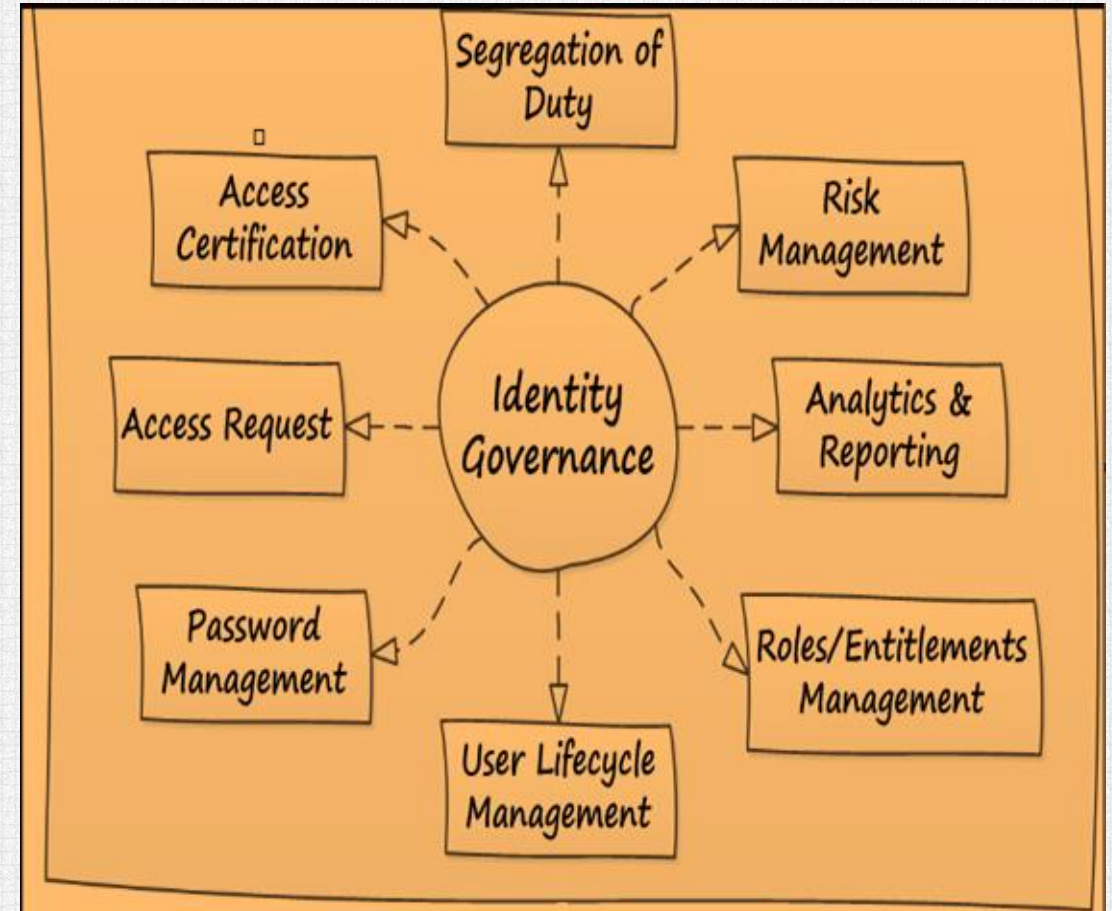
**Identity governance** is the policy-based centralized orchestration of user identity management and access control. Identity governance helps support enterprise IT security and regulatory compliance.

**Identity governance enables organizations to**

- Define
- Enforce
- Review and audit IAM policy, and also functions to compliance requirements and in turn audit user access to support compliance reporting.

**Identity governance will enhance the business performance and improves:**

- Access governance
- Data governance
- Privileged account governance
- Business-enabled access request and fulfillment
- Attestation or recertification
- Role engineering





# DIGITAL IDENTITY AND ACCESS MANAGEMENT SERVICES CONT..

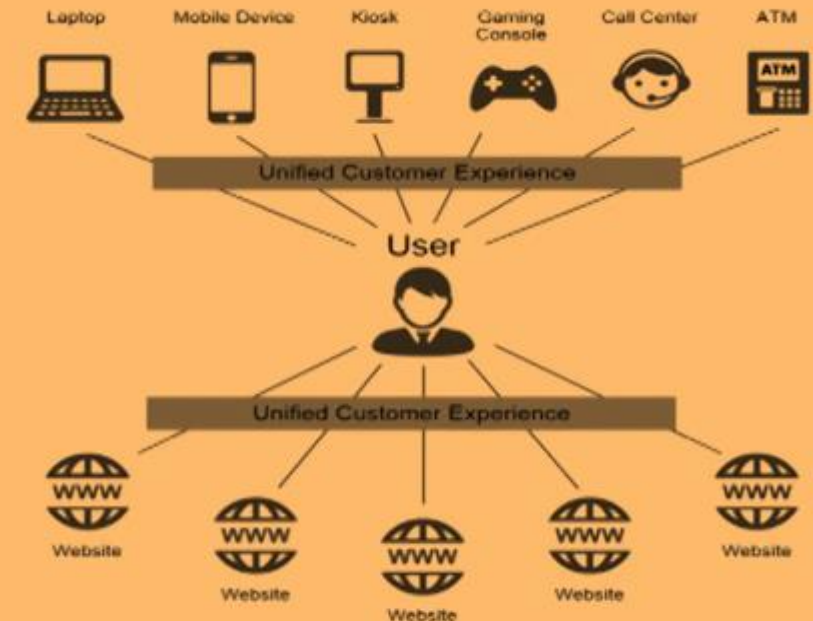
## Consumer Identity and access Management

Truly knowing the customer is the key to marketing success today. Not just knowing their name, but understanding their true identity. The service that makes this possible is known as customer identity and access management (CIAM).

These solutions help companies acquire and recognize customers, collect accurate first-party data, and facilitate personalized marketing through integrations

- Simplified Sing-on with social login
- Unified customer experience.
  - SSO across web application and other digital touch points
  - SSO across multiple accounts split by brand/company/product etc.
- Integrate the applications using the federation
- Universal logout

## Unify and Simplify the Customer Experience





# DIGITAL IDENTITY AND ACCESS MANAGEMENT SERVICES CONT..

**Mobile identity** is an extension and new innovative feature in digital identity. As enterprise adoption of mobility moves forward, the growth of the Internet of Things (IoT) presents the next evolution in computing. The impact of these trends in terms of increased risk to the business is straining existing security infrastructures

- Composes on concept of Any where, Any Mobile.
- Authentication remains at the core of information security in mobile identity.
- Authentication extends beyond user name and password, and in many instances the access token as well.
- Mobile identity also uses certification process to attest the authority of a particular individual to gain access to a specific resource from device to device network.
- Capable of provisioning and authenticate using identity attributes (i.e., biometric, thumb-prints) to proofs of private/secret tokens (cryptographic keys, passwords) and credentials and to location.





# DIGITAL IDENTITY AND ACCESS MANAGEMENT SERVICES CONT..

Cloud IAM solutions provide a clean and consistent access control interface for all cloud platform services. The same interface can be used for all cloud services. We can define roles and grant permissions to users to access resources at different granularity levels.

## Features

- Cloud IAM is referred to spend less on enterprise security by relying on the centralized trust model to deal with Identity Management across third-party and own applications.
- It enables your users to work from any location and any device.
- Provides access to all your applications using just one set of credentials through Single Sign-On.
- Add extra layers of security to your mission-critical apps using Multifactor Authentication.





# DIGITAL IDENTITY – SERVICES

## Biometrics and digital identity includes:

- Enhances the privacy, increase personal data control, and shift power relative to the monetization of consumer data.
- Biometric recognition are based on representations of biometrical data – biometrical characteristics using its own algorithm.

## Biometrics and digital identity includes:

- An integration platform mechanism focused on access management to identify the user and its access to the systems which supports formats like fingerprints, Facial Recognition, and Retina scan, DNA etc.
- Automated Identification solution deployed in sensitive areas like National Identification Centers, military/border, transportation etc.
- Strong Single Sign-On authentication mechanism to protect the system credentials, file encryptions, Emails, VPNs.





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**Q&A**



# Appendix

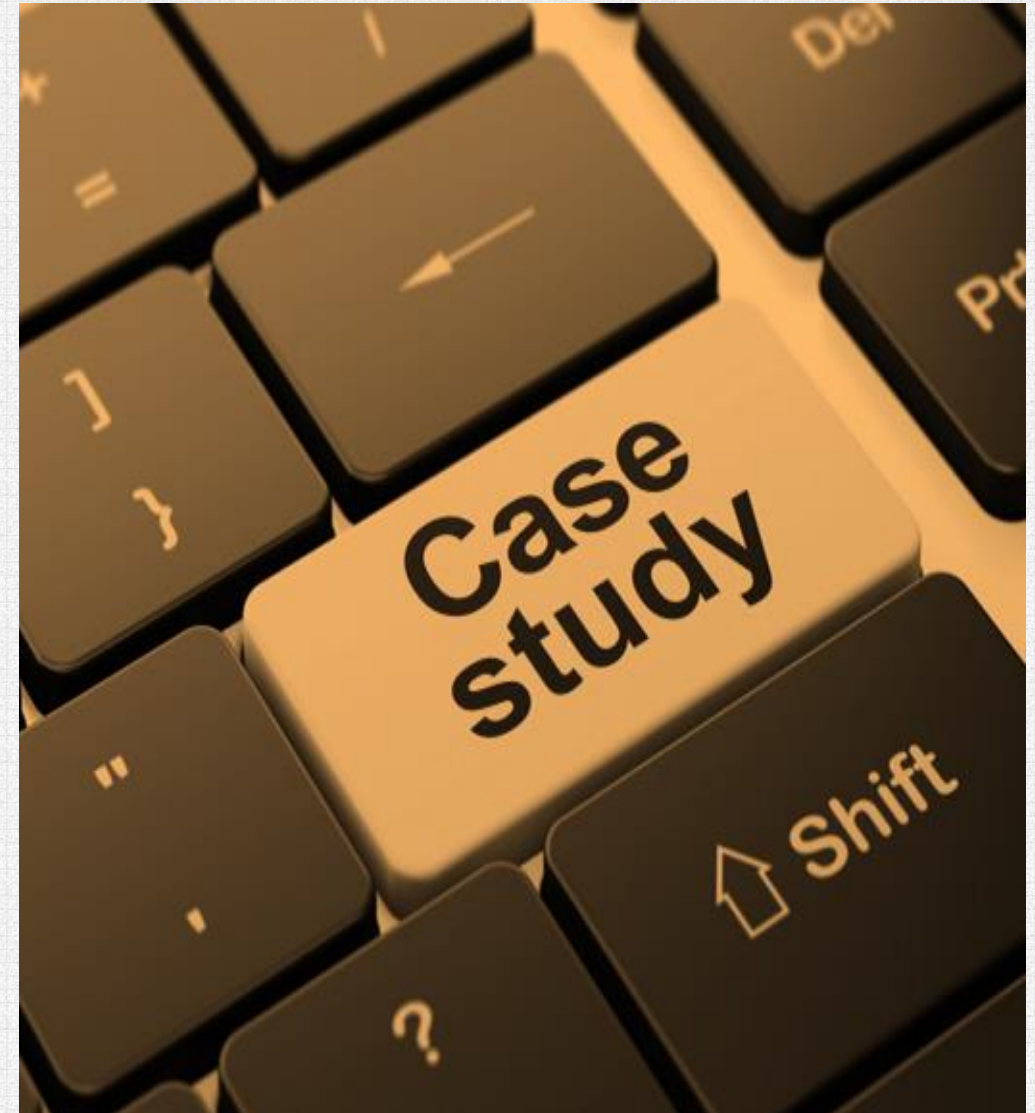


# CASE STUDY - TELIACOMPANY

- Client Name: TeliaCompany
- Operating Group: Telecom
- Geography: EALA
- TeliaCompany is the dominant telephone company and mobile network operator in Sweden and Finland. The company has operations in other countries in Northern and Eastern Europe, and in Central Asia and South Asia, with a total of 200 million mobile customers.

## What to Achieve

- Move the infrastructure to cloud
- Upgrade the OIM11g R1 to new version that is supported by vendor.
- Better user experience to the end customer.
- Attestation capabilities to address the regulatory requirements.





# CURRENT V/S FUTURE

## As-Is

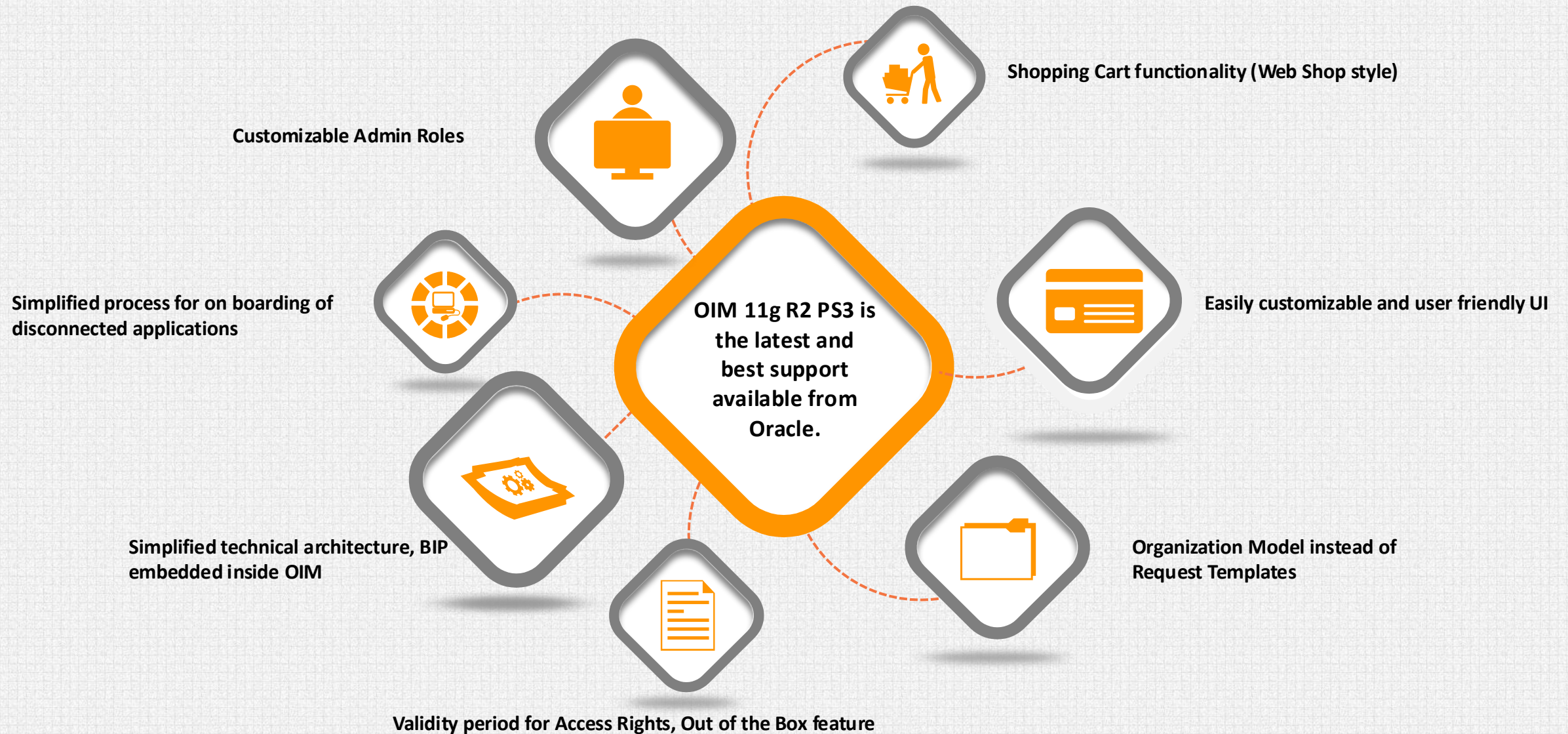
- The current IdM (OIM 11g R1) solution is reaching the end of its lifecycle so the solution should be replaced.
- The IdM solution is installed physical servers

## Current V/S future

## To-Be

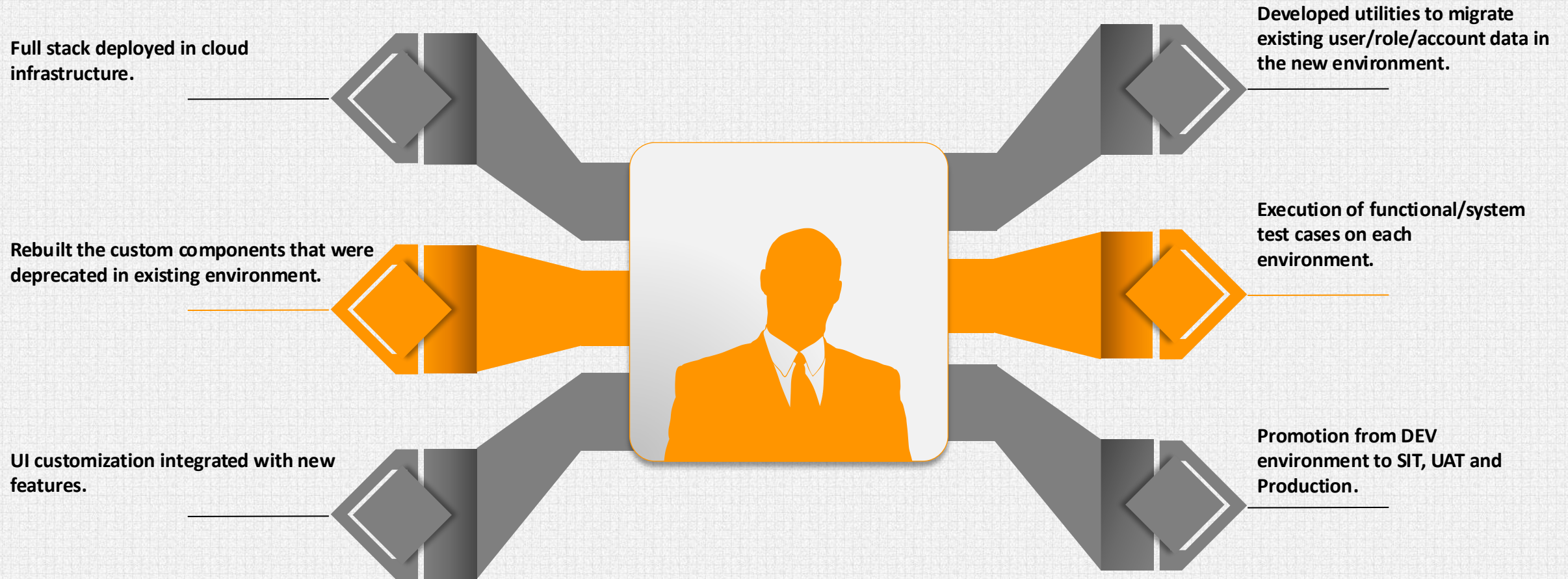
- Latest version of Oracle Identity Manager that supports cloud based deployments to be looked upon.
- Configure/Deploy the solution on-premises of cloud infrastructure
- Viable options of upgrade are
  - Upgrade to OIM 11g R2 PS3
  - Upgrade to OIM 11g R2 PS2

# SALIENT FEATURES OF 11G R2 PS3





# ACCENTURE ROLE





# BENEFITS/OUTCOMES

- Accenture completed the core project and extended to deliver additional capabilities. They were also involved in integrating the Identity Management with Access Management architectural changes to deliver auto-scaling solution that adapts to the current load. Improved quality of data allows greater insights in the audience.

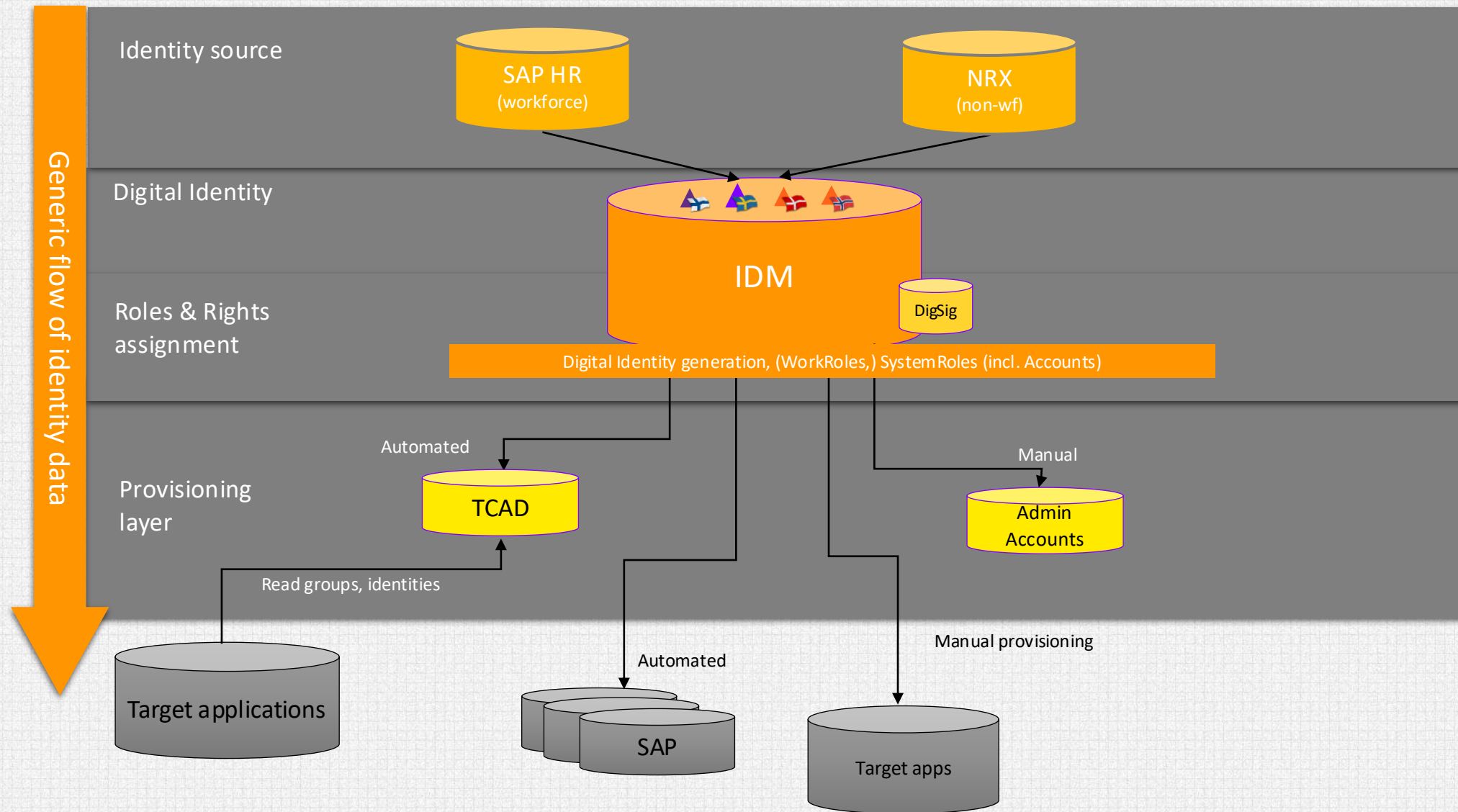
## Key benefits:

- New version OIM 11g R2 PS3 met all of the functional requirements proved scale through references and experience with Accenture.
- Strong platform architecture with Common Interfaces and Services.
- Deployment in Web-Services cloud environment (Vendor is Nimbus).
- More flexible and Agile when adapting to change.



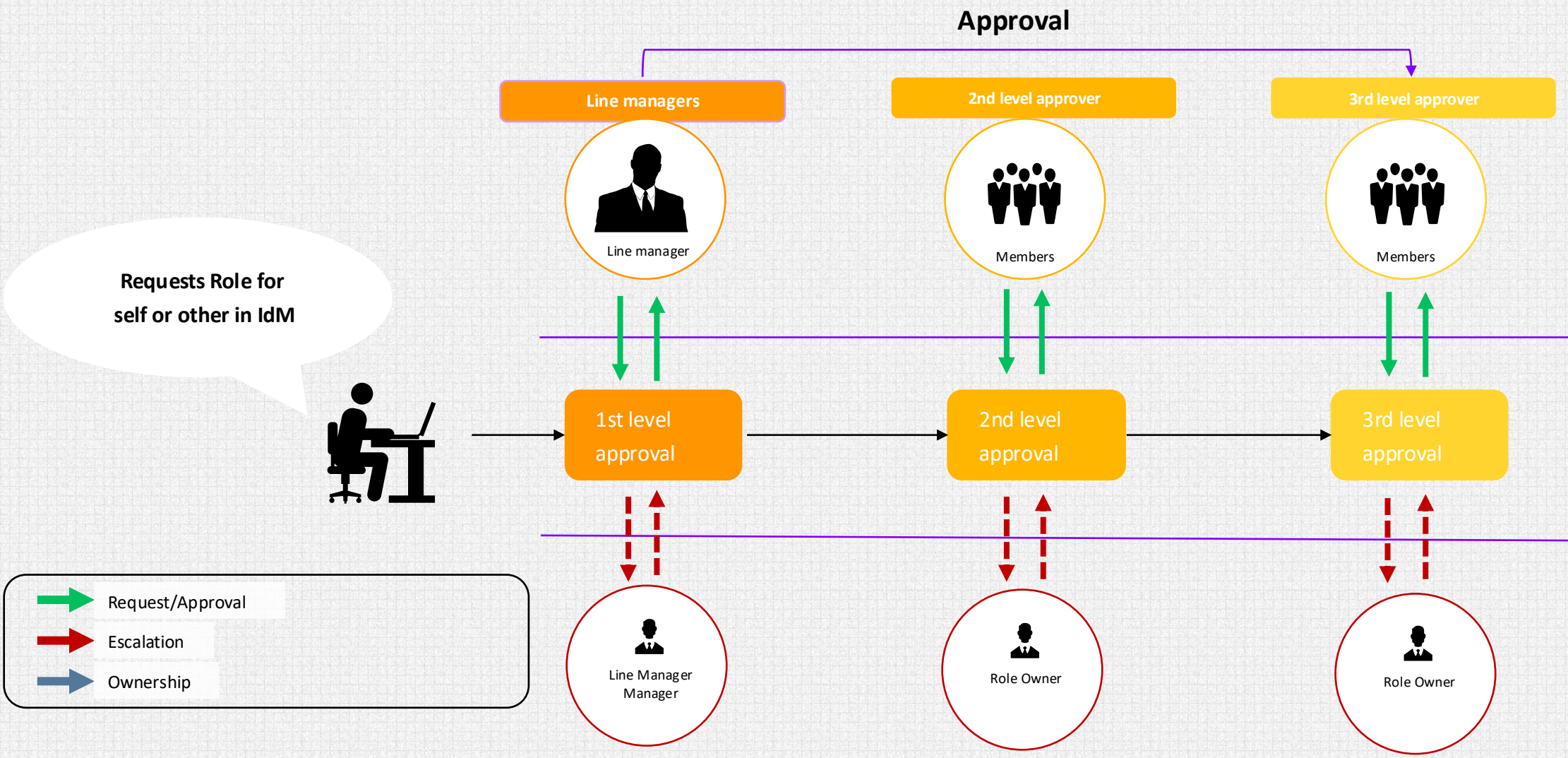


# TELIACOMPANY – IDM LANDSCAPE



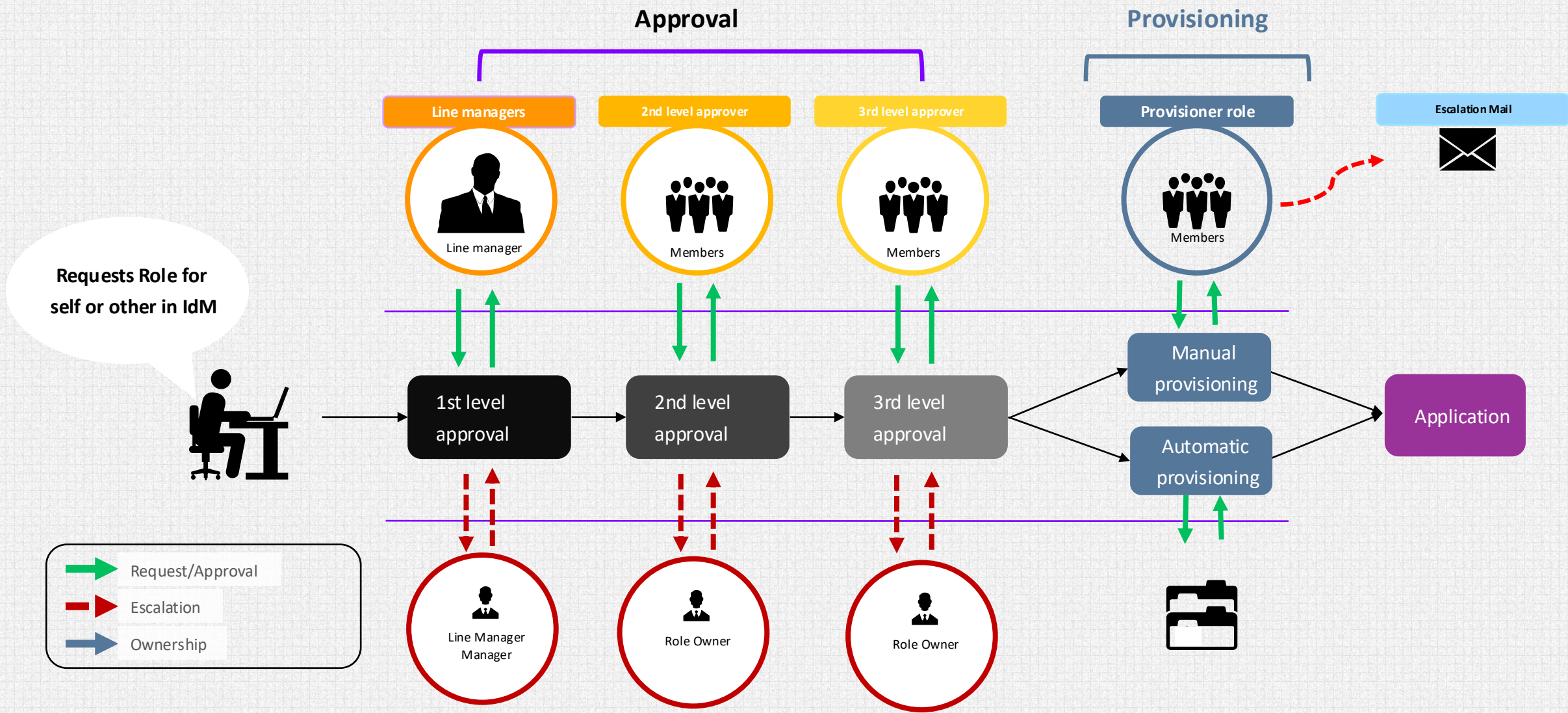


# CASE STUDY - ROLE REQUEST WORKFLOW



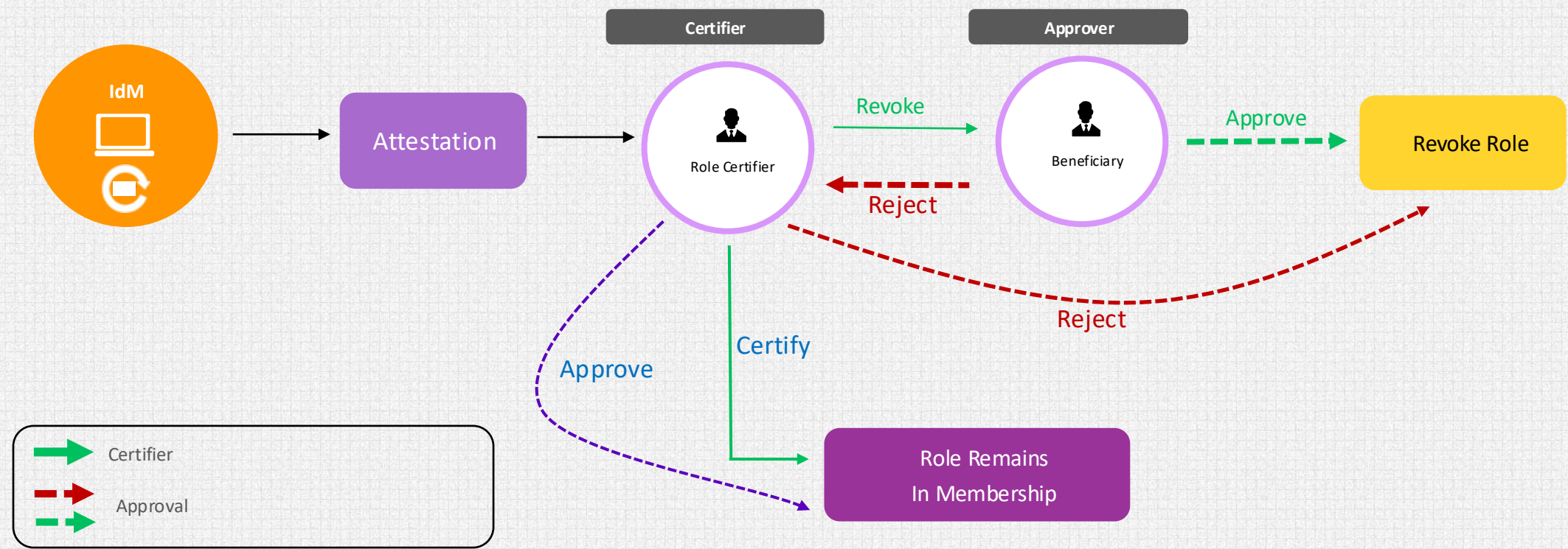


# CASE STUDY - ROLE REQUEST WORKFLOW





# CASE STUDY - ATTESTATION WORKFLOW





# DIGITAL IDENTITY - WHY

Documents or reports referred as sensitive, protected, confidential are stolen or erased without proper access are termed as breach.

Breaches mainly involves in personal healthcare information, credit card payments, personal identity theft.

Data breaches are not only particular to a person or a device, instead it is a victim of every organisation.

Data breach is extensively increased due to the increase in technology and insufficient measures.

## Features of digital identity are as follows:

- Provide a seamless secure experience for users across devices and communication channels
- Simple passwords is not acceptable
- Critical identity information should be efficiently managed in mobile, cloud services
- Enterprises should focus more on privileged accounts.







**THANK YOU**